Customer Behaviour and Repeat Purchases in Online Retail

Group name	**Ecom Analyst**
Dataset	UCI Online Retail, DOI: 10.24432/C5BW33
Code	**https://github.com/Sanskariiit-ai/ecommerce_analytics_cse558**

#	Name	Roll no.	Institute email	Contribution
1	**Sanskar**	**2024506**	sanskar24506@iiitd.ac.in	Dataset, analysis and code
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1. Dataset

Source: UCI Machine Learning Repository, Online Retail (Chen, 2015), https://doi.org/10.24432/C5BW33; CC BY 4.0.

Raw data: 541,909 rows and 8 original attributes; 1 December 2010–9 December 2011. Each row is an invoice product-line. The working audit adds one derived cancellation flag, so its intermediate schema has 9 columns; the prepared transaction table has additional derived fields.

Encoded-size check: Country has 38 levels in the encoded-size audit. One-hot encoding replaces the Country column with 38 indicator columns while retaining the other 7 original attributes: 7 + 38 = 45 features. Thus 541,909 × 45 = 24,385,905 encoded cells, above 3,000,000. If the derived cancellation flag is retained as a feature, there are 46 columns and 541,909 × 46 = 24,927,814 cells. Confirm the actual category count and encoded matrix shape when running the script; this arithmetic is not a substitute for the runtime shape check.

Feature	Type	Unit/range	Meaning and missingness
InvoiceNo	Nominal	Invoice ID	Invoice identifier; cancellation invoices often begin with C. No missingness reported in audit.
StockCode	Nominal	Product ID	Product identifier. No missingness reported in audit.
Description	Text	Product text	Product description; 1,454 rows missing.
Quantity	Discrete	Integer units	Units on line; 10,624 rows non-positive.
InvoiceDate	Datetime	2010-12-01 to 2011-12-09	Invoice timestamp.
UnitPrice	Continuous	GBP/unit	Unit price; 2,517 rows non-positive.
CustomerID	Nominal ID	Customer ID	Customer identifier; 135,080 rows missing.
Country	Nominal	Country labels	Customer country; encoded as one-hot indicators for the size check.

2. Data preparation

Issue	Action	Reason / alternative considered
Exact duplicate lines (5,268)	Remove exact duplicates	Prevent identical repeated records from inflating quantities and spend; retain non-identical lines.
Missing CustomerID (135,080)	Exclude from customer-level analysis; keep in raw audit	Imputing IDs would fabricate customer histories and repeat-purchase labels.
Missing descriptions (1,454)	Retain where other transaction fields are usable; description is not needed for customer-level target	Dropping all such rows would unnecessarily discard valid purchases.
Cancellation invoices (9,288)	Exclude from positive-sale cohort using invoice cancellation flag	Returns/cancellations should not count as positive purchases; still report them in audit.
Quantity ≤ 0 (10,624)	Exclude from positive-purchase analysis	Non-positive quantities represent returns/corrections rather than a positive purchase.
UnitPrice ≤ 0 (2,517)	Exclude from positive-purchase analysis	Non-positive prices do not represent positive sales value.
Skewed spend / outliers	Retain valid positive values; use log spend as an ML feature	High spend can be real customer behaviour. Automatic capping/deletion may erase genuine high-value customers.
Categorical country	One-hot encode in ML pipeline	Avoid imposing a false numeric order on country labels; unknown test categories are ignored by encoder.

3. What changed after preparation

Measure	Before	After
Rows	541,909 raw lines	392,692 positive-sale lines
Original columns	8	Prepared table has derived fields; customer table has 13 fields
Customers with IDs	4,372	4,338 customers after filtering
Distinct invoices	25,900 raw audit	18,532 positive-sale invoices
Exact duplicates	5,268	0 after exact duplicate removal
Missing CustomerID	135,080	Excluded from customer-level table, not imputed
Cancellation invoice lines	9,288	Excluded from positive-sale cohort
Non-positive quantity lines	10,624	Excluded from positive-sale cohort
Non-positive price lines	2,517	Excluded from positive-sale cohort

The prepared positive-sale table has 392,692 lines, 18,532 distinct invoices and 4,338 identified customers. These row counts are not additive because a row may meet more than one exclusion condition. The final customer table has one row per customer and supports inference without treating multiple invoices from one customer as independent observations.

4. Basic questions and problem statement

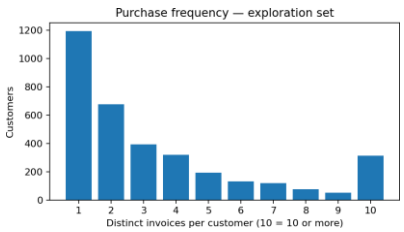


Figure 1: Purchase frequency in the exploration split is right-skewed: many customers have one invoice, with a long tail of repeat buyers.

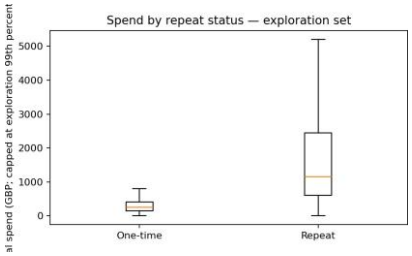


Figure 2: Exploration-split repeat customers show a broader total-spend distribution; display is capped at the exploration 99th percentile for readability.



Figure 3: Monthly invoice volume among exploration customers varies over time; December 2011 is incomplete because records end on 9 December.

Split before exploration

Customer-level data were split 80/20 with stratification on observed repeat-customer status and seed 558: exploration $n = 3,470$ and confirmation $n = 868$. EDA figures above use exploration customers only. The three tests below were run on the held-out confirmation customers; the median first-invoice threshold was set from the exploration split only (£300.655).

Obvious questions

1. How many customers buy once versus repeatedly?
2. Which customers spend the most over the observed period?
3. How does purchase activity vary by month?

Problem statements

P1 (primary). Is the probability of repeat purchasing different for customers whose first invoice value is above the exploration-set median compared with those at or below it?

P2. Does customer-level mean invoice value differ between repeat and one-time customers?

P3. Does repeat-purchase probability differ between UK and non-UK customers?

Novelty: The obvious questions describe historical volume and spend. These statements test pre-specified customer-level contrasts in a held-out confirmation sample, use a future-relevant retention outcome for P1, and explicitly avoid counting each invoice as an independent customer. They remain observational associations, not causal claims.

5. Statistical inferences

All tests use the confirmation set (868 customers), two-sided alternatives, and $\alpha = 0.05$. Practical thresholds below are chosen for interpretation in this report and were not preregistered; therefore practical-significance conclusions are exploratory. For two proportions, the minimum practically meaningful difference is 5 percentage points. For average invoice value, it is £50 per customer.

Test 1 (primary): first-invoice spend and repeat purchasing

a) Hypotheses and method

H_0	$p_{above} - p_{at/below} = 0$
H_1	$p_{above} - p_{at/below} \neq 0$
Significance	$\alpha = 0.05$; two-sided
Test	Two-proportion z-test
Practical threshold	5 percentage points

The test compares independent customers in the two first-invoice groups. A two-proportion z-test is suitable for the difference in proportions at these sample sizes.

b) Other tests considered

A chi-square test of independence on the corresponding 2 × 2 table is asymptotically equivalent and would give the same two-sided conclusion. Fisher's exact test avoids the large-sample approximation but is not necessary with these counts. A one-sided test was not used because the question allows either direction.

c) Result

Above-median group: 290/424 = 68.40%; at/below-median: 279/444 = 62.84%. Difference = 5.56 percentage points; 95% CI [-0.75, 11.87] percentage points; $z = 1.723$, degrees of freedom: not applicable for a z-test, $p = 0.085$. Fail to reject H_0. The point estimate is just above the 5-point practical threshold, but the interval includes zero and values below the threshold; evidence is inconclusive.

Test 2: customer-level average invoice value by repeat status

a) Hypotheses and method

H_0	$\mu_{repeat} - \mu_{one-time} = 0$
H_1	$\mu_{repeat} - \mu_{one-time} \neq 0$
Significance	$\alpha = 0.05$; two-sided
Test	Welch two-sample t-test
Practical threshold	£50 per customer

Each customer contributes one average invoice value. Welch's test does not require equal population variances.

b) Other tests considered

The pooled t-test assumes equal variances; we avoid that assumption. A Mann–Whitney test compares ranks/distributions, not specifically means. A bootstrap confidence interval would be a useful robustness check for the skewed values. A test on invoice-level rows was rejected because repeat customers contribute multiple invoices, creating pseudoreplication.

c) Result

Repeat customers: mean £398.95 ($n = 569$); one-time customers: £384.08 ($n = 299$). Difference = £14.88; 95% CI [-£54.48, £84.23]; $t = 0.421$, $df = 530.68$, $p = 0.674$. Fail to reject H_0. The point estimate is below £50, but the interval includes differences larger than the practical threshold in either direction; the data do not establish equivalence.

Test 3: UK versus non-UK repeat-purchase rates

a) Hypotheses and method

H_0	$p_{UK} - p_{nonUK} = 0$
H_1	$p_{UK} - p_{nonUK} \neq 0$
Significance	$\alpha = 0.05$; two-sided
Test	Two-proportion z-test
Practical threshold	5 percentage points

b) Other tests considered

A chi-square test of independence is asymptotically equivalent. Fisher's exact test is a possible small-cell alternative, although the non-UK group has 95 customers and observed success/failure counts are adequate for the normal approximation. A logistic regression adjusted for country/customer covariates could address confounding but answers a different, adjusted question.

c) Result

UK repeat rate: 508/773 = 65.72%; non-UK: 61/95 = 64.21%. Difference = 1.51 percentage points; 95% CI [-8.70, 11.71] percentage points; $z = 0.292$, degrees of freedom: not applicable for a z-test, $p = 0.770$. Fail to reject H_0. The interval is wide and does not rule out practically meaningful differences.

6. Scalable machine learning

The ML target is whether a customer with purchase history before 1 September 2011 purchases during the next 90 days. Features are calculated only from the history period. The cohort contains 3,317 customers; 1,864 (56.2%) have a positive label. On a fixed 830-customer test set, training size was varied across 621, 1,243 and 2,487 customers using the same seed, preprocessing and model settings.

Model	Train n	F1	ROC-AUC	Fit (s)
Logistic regression	621	0.671	0.718	0.016
Logistic regression	1,243	0.686	0.726	0.016
Logistic regression	2,487	0.693	0.732	0.027
Shallow tree	621	0.636	0.701	0.009
Shallow tree	1,243	0.667	0.717	0.032
Shallow tree	2,487	0.623	0.727	0.039

Logistic regression achieves the best full-sample F1 and ROC-AUC. Tree performance is not monotonic with training size. Fit times are machine-specific and this small customer-level dataset does not prove large-scale scalability; the experiment is a controlled sample-size comparison. The evaluation split is stratified randomly within a temporal cohort, not a pure later-period holdout, which limits deployment claims.

7. Limitations

- **Coverage.** One UK retailer over roughly one year; December 2011 is incomplete.
- **Missing customer IDs.** 135,080 raw rows cannot be reliably linked to customer histories and are excluded from customer-level analyses.
- **Practical thresholds.** Thresholds were selected for interpretation in this report, not preregistered before analysis.
- **Association, not causation.** Spending, geography and repeat status may be related to unobserved customer characteristics.
- **Encoding verification.** The size arithmetic uses 38 observed Country levels from the prior audit; the exact encoded matrix shape must be verified by running the included script on the original raw file.